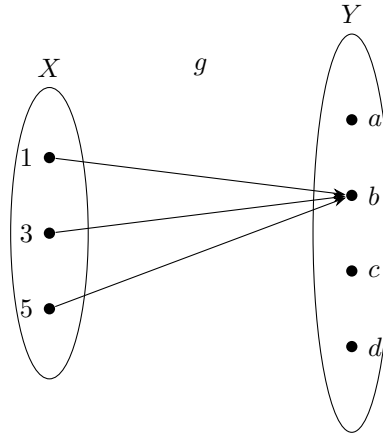


Weekly Assignment: Functions (Epp Ch. 7)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 7.1 #2). Let $X = \{1, 3, 5\}$ and $Y = \{a, b, c, d\}$. Define $g: X \rightarrow Y$ by the following arrow diagram.

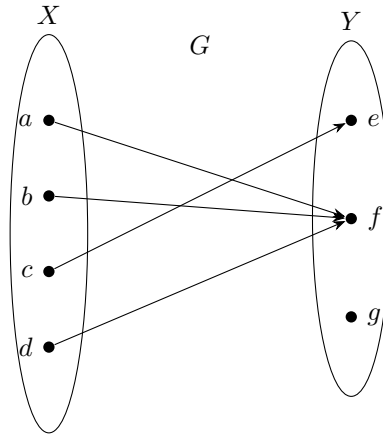


- Write the domain of g and the co-domain of g .
- Find $g(1)$, $g(3)$, and $g(5)$.
- What is the range of g ?
- Is 3 an inverse image of a ? Is 1 an inverse image of b ?
- What is the inverse image of b ? Of c ?
- Represent g as a set of ordered pairs.

Problem 2 (Epp 7.1 #4).

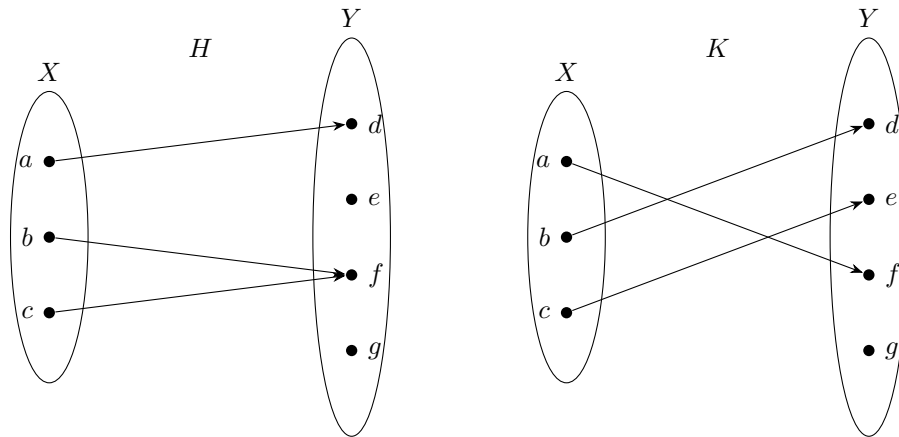
- Find all functions from $X = \{a, b\}$ to $Y = \{u, v\}$.
- Find all functions from $X = \{a, b, c\}$ to $Y = \{u\}$.
- Find all functions from $X = \{a, b, c\}$ to $Y = \{u, v\}$.

Problem 3 (Epp 7.2 #7b). Let $X = \{a, b, c, d\}$ and $Y = \{e, f, g\}$. Define the function $G: X \rightarrow Y$ by the following arrow diagram.



Is G one-to-one? Why or why not? Is G onto? Why or why not?

Problem 4 (Epp 7.2 #8). Let $X = \{a, b, c\}$ and $Y = \{d, e, f, g\}$. Define functions H and K by the following arrow diagrams.



a. Is H one-to-one? Why or why not? Is it onto? Why or why not?

b. Is K one-to-one? Why or why not? Is it onto? Why or why not?

Problem 5 (Epp 7.3 #5). Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by the rule $f(x) = 2x$ for every real number x . Find $(f \circ f)(x)$.

Problem 6 (Epp 7.3 #11). Define $F: \mathbb{R} \rightarrow \mathbb{R}$ and $G: \mathbb{R} \rightarrow \mathbb{R}$ by the rules

$$F(x) = 3x \quad \text{and} \quad G(x) = \left\lceil \frac{x}{3} \right\rceil \quad \text{for every real number } x.$$

a. Find $(G \circ F)(6)$, $(F \circ G)(6)$, $(G \circ F)(1)$, and $(F \circ G)(1)$.

b. Is $G \circ F = F \circ G$? Explain.

Problem 7. Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$. Prove or disprove: if $g \circ f$ is bijective, then f is bijective.